

Task 1:

Comparison Table of DDR3, DDR4, and DDR5 RAM

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
Data Speed	800 MHz to 2133 MHz	1600 MHz to 3200 MHz	4800 MHz to 8400+ MHz
Operating Voltage	1.5V (Higher power consumption)	1.2V (More energy efficient)	1.1V (Most power efficient)
Maximum Stick Capacity	Up to 8 GB per module	Up to 32 GB per module	Up to 128 GB+ per module
Architecture	Single-channel sub-channel	Single 64-bit channel per stick.	Dual 32-bit independent sub-

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
	architecture.		channels per stick.
Typical Use Cases	Older legacy systems, basic office work, and budget laptops.	Standard modern gaming, video editing, and mainstream business PCs.	High-end gaming rigs, 4K/8K video rendering, and heavy server workloads.

Task 2:

RAM Identification and Reinstallation (Video Reference)

- **Since I am unable to open my current device due to tool constraints, I am sharing a trusted tutorial video that demonstrates the exact process of locating, removing, and installing a RAM module in a laptop.**

Reference Video Link: [How to install memory in a laptop PC \(Kingston Technology\)](#)

Key steps observed in the process:

- 1. Shut down the system and disconnect the power supply.**

- 2. Open the back panel to locate the RAM slots on the motherboard.**
- 3. Gently pull the side clips apart to release the current RAM stick.**
- 4. Align the notch on the new RAM stick with the slot, insert it at an angle, and press down until the side clips snap securely into place.**
- 5. Reassemble the device and boot it up to verify the new memory is recognized.**

Task 3:

Comparison of HDD, SSD, and NVMe Storage Types

Feature / Difference	HDD (Hard Disk Drive)	SSD (SATA Solid State Drive)	NVMe (Non-Volatile Memory Express)
Technology / Mechanism	Uses moving mechanical parts (spinning magnetic platters and a read/write head).	Uses flash memory chips (No moving parts).	Uses high-speed flash memory chips directly connected via PCIe.
Data Transfer Speed	Slower speeds (typically around 100-150 MB/s).	Faster speeds (typically around 500-550 MB/s).	Ultra-fast speeds (can reach 3500-7000+ MB/s).

Feature / Difference	HDD (Hard Disk Drive)	SSD (SATA Solid State Drive)	NVMe (Non-Volatile Memory Express)
Form Factor & Connection	Bulky 3.5" or 2.5" drive connected via standard SATA cables.	Standard 2.5" drive shape connected via SATA cables.	Small, slim stick (M.2 form factor) mounted directly onto the motherboard.

Task 4:

Storage Drive Identification and Reinstallation (Video Reference)

- **Since I do not have a spare device that can be safely disassembled, I am fulfilling this task's constraint by providing a trusted tutorial video that demonstrates how to locate, remove, and reinstall an SSD/HDD, along with identifying its connector type.**

Reference Video Link:

[How to Install an NVMe or SATA M.2 SSD in a Laptop/PC](#)

Key steps observed for storage drive removal and installation:

- 1. Disconnect the power supply and remove the bottom panel of the laptop/PC.**

- 2. Locate the storage drive. If it is a traditional 2.5" drive, it will use a standard SATA connector. If it is a small stick-like drive, it will use an M.2 / NVMe connector directly on the motherboard.**
- 3. Unscrew the mounting bracket or the single M.2 holding screw.**
- 4. Gently slide the drive out of the connector socket at a slight angle.**
- 5. To reinstall, align the notch of the SSD with the motherboard slot, insert it firmly, push it down flat, and tighten the holding screw.**
- 6. Reassemble the device cover and boot into the BIOS or Windows Disk Management to verify the storage drive is successfully recognized.**